

Yagyaansh Khaneja

contact@yagyaansh.com
github.com/YagyaanshK
[LinkedIn](#) | [X](#) | [YouTube](#)

Education

IIT Roorkee, B. Tech. in Engineering Physics (CGPA 7.11/10)	2021
RDPS, Delhi (CBSE), Class XII: PCM (94.2 %)	2017
RDPS, Delhi (CBSE), Class X (CGPA 10/10)	2015

Skills

- Programming:** Python, FORTRAN, HTML/CSS, SQLite
- Tools:** MATLAB, COMSOL, CST MW Studio, Excel/Word/PowerPoint, LaTeX, Lightroom
- Languages:** English (Native), Hindi (Native), French (Basic), Punjabi (Basic)

Certificates

- Machine Learning:** Andrew Ng ([Coursera](#)) [2023]
- Python for Everybody:** Charles Severance ([Coursera](#) Specialization) [2020]
- Web Design for Everybody:** Colleen van Lent ([Coursera](#) Specialization) [2020]
- Internet: History, Technology, Security:** Charles Severance ([Coursera](#)) [2020]

Achievements

- NGPE 2019 (Part A):** Top 10%
- JEE 2017 (Advanced):** AIR 3188
- JEE 2017 (Mains):** AIR 2933 (Top 0.2%)
- Vijyoshi 2016 (Camp):** Invited Guest
- KYPY 2015 (SA):** AIR 300 (Top 0.4%)
- NTSE 2015 (II):** Scholar (Top 0.1%)

Miscellaneous

- STEM & English Teacher** (May-Aug, 2022)
 - Locations:** Ladakh ([SECMOL](#)), Vietnam ([Workaway](#))
 - Taught dropouts using hands-on science/math [experiments](#).
 - Led astronomy and photography hobby clubs.
 - Skills:** Communication, Empathy
- Leadership Roles, IIT Roorkee** (2017-2021)
 - Additional Secretary, Photography Section
 - Coordinator, Thomso (Cultural Fest)
 - Content Manager, Srishti (Tech Exhibition)
 - Skills:** Leadership, Management

Work Experience:

- **Project Manager (Volunteer) – Tech Team** [Dec 2024 – Jun 2025]
 - Organizations involved:** [Tarayana](#), [GIAOS](#), [Omniverse](#)
 - Work:** Guided the development a [data dashboard](#) (proof-of-concept) for a remote village in Bhutan (tech stack: React Native, DeepSeek, MongoDB)
 - Skills:** Team Management, High Level Architecture Design, Python
- **Family Business – B2B Wholesale (pan India)** [Sep 2022 – present]
 - Location:** Asia's largest wholesale market: Chandni Chowk/Chawri Bazaar (Delhi-6)
 - TL;DR:** In essence, increased the efficiency of operations, **saving 4 hours per day**
 - Migrated business **accounts** from legacy systems to cloud accounting software, enabling remote work for the first time in 35 years.
 - Meticulously classified 1000s of **Steel products**, 400+ parties; making data entry organized & eliminating the need for “data cleaning” step (during data analysis) [working on Data Entry automation currently]
 - Streamlined packaging and dispatch process in warehouse using stickers, labels, item codes, and colour coding for dispatch details.
 - Started **online retail** sales channel through [Meesho](#).
 - Designed a [catalog](#) of Door Fittings, Fasteners, Screws - simplifying products for customers
 - Learned invaluable lessons about Money, **Cash Flow**, Relationship Management, Business Ethics, Sales, etc.
 - Skills:** Warehouse Management, Canva Design, Accounting, Financial Management
- **Research Intern (Energy Team) – Sonam Wangchuk** [Jan 2022 – May 2022]
 - Organization:** Himalayan Institute of Alternatives ([HIAL](#)), Ladakh
 - TL;DR:** In essence, assisted [Sonam](#) on his Sustainable Development projects
 - Drafted training [manual](#) for construction & working principles of Passive Solar Heated ([PSH](#)) Buildings in Ladakh.
 - Installed new 7.5 kW Solar PV; maintained existing 20 kW Solar PV and 240V (DC to AC) Lead Acid battery house.
 - Installed new Stevenson Screen; assisted in weather [data collection](#) (& analysis) in remote regions of Ladakh using SensorPush and Campbell Scientific weather stations.
 - Energy Analysis of Phase Change Materials (PCMs) in PSH Trombé Walls, and HVAC systems
 - Skills:** Research (Thermodynamics), MS Excel, Project Implementation, Electrical

Physics Projects

- **Bachelor's Project – Silicon Photonic Switch, IIT Roorkee** [Dec 2018 – Sep 2021]
 - TL;DR:** Simulated high-performance photonic switches (GST-SOI waveguides) using CST MW Studio and COMSOL.
 - Designed non-volatile switches for optical communication at 2.1µm
 - Studied Phased Array Waveguides & WDM for applications in data communication
 - Skills:** EM Simulation ([COMSOL](#), CST MW Studio), MATLAB, Waveguide Physics
- **Research Camp – Optics/Photonics, ITMO University** [May 2019 – Jun 2019]
 - Organization:** ITMO University, St. Petersburg, Russia
 - TL;DR:** Gained [exposure](#) to advances in Optics through active engagement with frontier research
 - Generated THz pulse radiation using laser filamentation of water.
 - Experimented with liquids of varying pH levels, to study the effect on intensity of THz generation (for the first time in the world!)
 - Studied Quantum Information, Holography, QKD Systems, Femtosecond Lasers, etc.
 - Skills:** Experimental Physics, Paper Presentation, Laser Operation

Other Cool Stuff

- **Personal Website** – vagyaansh.com [Nov 2020 – Jan 2021]
- TL;DR: Hard-coded my website using basic HTML, CSS and JavaScript (no frameworks)
 - Implemented Dark Theme toggle using JavaScript
 - Wrote blogs with graphics, integrated comments section using Disqus
 - Hosted my website using GitHub Pages
- Skills: Web Development (HTML, CSS), handling SVG in HTML

- **Shady Stuff – Fee Data Extraction** [Aug 2020]
- TL;DR: Made API calls (Azure server) to extract college fee receipts (sensitive data of 3000+ students including bank name and transaction id)
 - Libraries used - urllib.request, ssl, json, sqlite3 in Python
 - converted the received json data into a SQLite database for easy accessibility
- Skills: Python, Handling JSON and SQLite database using Python